Mass Transfer

Equations of Change for Multicomponent Systems

Volume element Δx, Δy, Δz fixed in space through which a binary mixture of A and B is flowing. Withing the element, A may be produced by a chemical reaction at a rate of r_A $(M/(L^3t))$.

Rate of change of mass of A in the volume element: $\dfrac{\partial \rho_A}{\partial t}\Delta x\,\Delta y\,\Delta z$

Input of A across face at x: $n_{Ax}\big|_x\,\Delta y\,\Delta z$

Output of A across face at x+ Δx: $n_{Ax}\big|_{x+\Delta x}\,\Delta y\,\Delta z$

Rate of production of A by chemical reaction. $r_A\,\Delta x\,\Delta y\,\Delta z$

n_A is the mass flux of A:

Considering the input and output terms in the y and z directions, the entire mass balance equation can be written as

$$\frac{\partial \rho_A}{\partial t} + \left(\frac{\partial n_{Ax}}{\partial x} + \frac{\partial n_{Ay}}{\partial y} + \frac{\partial n_{Az}}{\partial z} \right) = r_A$$

Equation of continuity of species A

This represents the change in mass concentration of A w.r.t. time at a fixed point in space, the change resulting from motion of A and chemical reactions producing A

$$\frac{\partial \rho_A}{\partial t} + \left(\nabla . n_A \right) = r_A$$

$$\frac{\partial \rho_B}{\partial t} + \left(\nabla . n_B \right) = r_B$$

$$\frac{\partial \rho}{\partial t} + \left(\nabla . \rho v \right) = 0$$

In order to obtain the concentration profiles, the fluxes n_A and N_A are to be replaced by appropriate expressions involving concentration gradients

$$n_A - w_A\left(n_A + n_B\right) = -\rho D_{AB} \nabla w_A$$ Relation involving mass fraction w_A

$$N_A - x_A\left(N_A + N_B\right) = -C D_{AB} \nabla x_A$$ Relation involving mole fraction x_A

Substituting in the equation of continuity in the respective cases, the diffusion equation is obtained

$$\frac{\partial \rho_A}{\partial t} + \left(\nabla . \rho_A v\right) = \left(\nabla . \rho D_{AB} \nabla w_A\right) + r_A$$ $$w_A\left(n_A + n_B\right) = w_A \rho v = \rho_A v$$

$$\frac{\partial \rho_A}{\partial t} + \left(\nabla . \rho_A v\right) = \left(\nabla . \rho D_{AB} \nabla w_A\right) + r_A$$

Special Case: Constant ρ and D_{AB}

$$\frac{\partial \rho_A}{\partial t} + \rho_A \left(\nabla . v\right) + v . \left(\nabla \rho_A\right) = D_{AB} \nabla^2 \rho_A + r_A$$

$$\frac{\partial \rho_A}{\partial t} + v . \left(\nabla \rho_A\right) = D_{AB} \nabla^2 \rho_A + r_A$$

Dividing by M_A

$$\boxed{\frac{\partial C_A}{\partial t} + v . \left(\nabla C_A\right) = D_{AB} \nabla^2 C_A + R_A}$$

Mass Transfer Fundamentals

BSL

Fick's Law $N_A = x_A\,(N_A + N_B) - \boxed{D_{AB}\nabla C_A}$

Molar Flux Bulk Flow Diffusion

$$D_{AB}\left(\frac{\partial C_A}{\partial x} + \frac{\partial C_A}{\partial y} + \frac{\partial C_A}{\partial z}\right)$$

Species Balance Equation in Cartesian Coordinate System

$$\left(\frac{\partial C_A}{\partial t} + v_x\frac{\partial C_A}{\partial x} + v_y\frac{\partial C_A}{\partial y} + v_z\frac{\partial C_A}{\partial z}\right) = D_{AB}\left[\frac{\partial^2 C_A}{\partial x^2} + \frac{\partial^2 C_A}{\partial y^2} + \frac{\partial^2 C_A}{\partial z^2}\right] + R_A$$

Transient, Convection, Conduction/Diffusion, Generation

$$\rho\hat{C}_p\left(\frac{\partial T}{\partial t} + v_x\frac{\partial T}{\partial x} + v_y\frac{\partial T}{\partial y} + v_z\frac{\partial T}{\partial z}\right) = k\left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}\right] + \mu\phi_v + \dot{Q}$$

Observations

$$\left(\frac{\partial C_A}{\partial t} + v_x \frac{\partial C_A}{\partial x} + v_y \frac{\partial C_A}{\partial y} + v_z \frac{\partial C_A}{\partial z}\right) = D_{AB}\left[\frac{\partial^2 C_A}{\partial x^2} + \frac{\partial^2 C_A}{\partial y^2} + \frac{\partial^2 C_A}{\partial z^2}\right] + R_A$$

Homogeneous reaction term appears in the governing equation

Heterogeneous reaction appears as a boundary condition.

Possible boundary conditions

1. Concentration specified at a point e.g. solubility, equilibrium etc.

2. At the catalyst surface, diffusive flux to the surface must be equal to the rate of reaction

$$-D_{AB}\frac{dC_A}{dx}\Big|_{x=\text{cat sur}} = -R_A^{//}$$

$^{//}$ refers to heterogeneous reaction

Boundary conditions contd.

3. Equality of concentration at the interface $C_{A,\,s1} = C_{A,\,s2}$

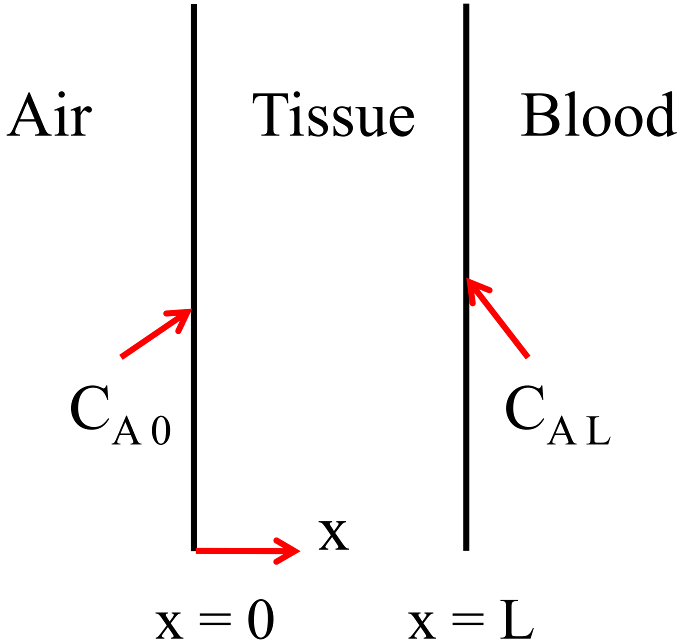
4. Impervious surface $dC/dx = 0$ at specified location

5. Conductive flux to the solid-fluid interface is equal to the convective flux away from the interface, $-D_{AB}\ dC_A/dx\,\big|_{x=L} = h_m\,(C_{As} - C_{A\infty})$

Problem

Consider the problem of oxygen transfer from the interior lung cavity, across the lung tissue, to the network of blood vessels on the opposite side. The lung tissue (species B) may be approximated as a plane wall of thickness L. The inhalation process may be assumed to maintain a constant molar concentration $C_A(0)$ of oxygen (species A) in the tissue at its inner surface ($x = 0$), and assimilation of oxygen by the blood may be assumed to maintain a constant molar concentration $C_A(L)$ of oxygen in the tissue at its outer surface ($x = L$). There is oxygen consumption in the tissue due to metabolic processes, and the reaction is zero order, with rate constant equal to k_0'''.

Obtain expressions for the distribution of oxygen concentration in the tissue and for the rate of assimilation of oxygen by the blood per unit tissue surface area.



Observations

1. Oxygen assimilation in the tissue = k_0'''
2. Homogeneous reaction, $R_A = -k_0'''$
3. diffusion only reaction, no convection
4. Steady state
5. $C_A = f(x)$ only

$$\left(\cancel{\frac{\partial C_A}{\partial t}} + v_x \cancel{\frac{\partial C_A}{\partial x}} + v_y \cancel{\frac{\partial C_A}{\partial y}} + v_z \cancel{\frac{\partial C_A}{\partial z}}\right) = D_{AB}\left[\frac{\partial^2 C_A}{\partial x^2} + \cancel{\frac{\partial^2 C_A}{\partial y^2}} + \cancel{\frac{\partial^2 C_A}{\partial z^2}}\right] + R_A$$

Obs. 4 3 3 3 5 5 2

Gov. Eqn $$D_{AB}\frac{d^2 C_A}{dx^2} - k_0''' = 0$$

$$D_{AB}\frac{d^2 C_A}{dx^2} - k_0''' = 0$$

$$C_A = \frac{k_0'''}{2\, D_{AB}}\, x^2 + C_1\, x + C_2$$

Boundary Conditions

$C_A = C_{A0}$ at $x = 0$ $\longrightarrow$ $C_2 = C_{A0}$

$C = C_{AL}$ at $x = L$ $\longrightarrow$ $C_1 = \frac{C_{AL} - C_{A0}}{L} - \frac{k_0 L}{2\, D_{AB}}$

Therefore, the concentration distribution of O_2 in the lung tissue is

$$C_A\,(x) = \frac{k_0'''}{2\,D_{AB}}\,x\,(\,x - L) + \,[C_{AL}\; - \;C_{A0}]\,\frac{x}{L} + \,C_{A0}$$

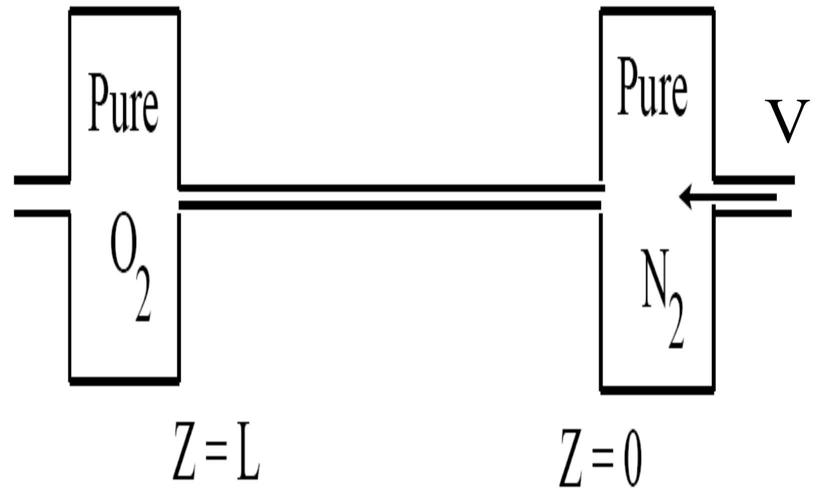
The oxygen assimilation rate per unit area is given by

$$N_A = -\,D_{AB}\,\frac{dC_A}{dx}\,\bigg|_{x=L} = -\,\frac{k_0 L}{2} + \frac{D_{AB}}{L}\,[C_{A0}\; - \;C_{AL}]$$

HOW?

Problem

Two gases (O_2 and N_2) are separated into two bulbs connected by a narrow diameter capillary. Pure N_2 flows from right to left (with a velocity v), as shown in the figure. To what degree can O_2 diffuse upstream against the N_2 flow, i.e, derive the concentration distribution of O_2 in the tube.



Assume that the system has achieved some steady state and that the volumes at the two ends are so large that each remains nearly pure, even though some exchange takes place (early stage steady state model.

Find the solution for the limiting case of small vL/D, where D is the diffusion coefficient.

Species Balance Equation

Non ~~zero~~

$$\left(\cancel{\frac{\partial C_A}{\partial t}} + \cancel{v_x \frac{\partial C_A}{\partial x}} + \cancel{v_y \frac{\partial C_A}{\partial y}} + v_z \frac{\partial C_A}{\partial z}\right) = D_{AB}\left[\cancel{\frac{\partial^2 C_A}{\partial x^2}} + \cancel{\frac{\partial^2 C_A}{\partial y^2}} + \frac{\partial^2 C_A}{\partial z^2}\right] + \cancel{R_A}$$

Observations (Component A represents O_2)

1. No reaction
2. The volumes at the two ends are large such that each remains pure
3. C_A is a function of z only
4. Steady state
5. v_x and v_y are zero, v_z in nonzero

Therefore, the governing equation becomes

$$v\frac{dC_A}{dz} = D_{AB}\frac{d^2C}{dz^2}$$

The boundary conditions are

$C_A = C_{A0}$ at $Z = L$, $C_A = 0$ at $Z = 0$

Let $\dfrac{dC_A}{dt} = p$ $\qquad \dfrac{dp}{dz} - \dfrac{v}{D_{AB}}\, p = 0$

$p = a'\, e^{vz/D_{AB}}$ a' being the integration constant

Therefore, $$C_A = a\, e^{vz/D_{AB}} + b$$

Using the boundary conditions

$C_A = C_{A0}$ at $Z = L$, $C_A = 0$ at $Z = 0$

a and b can be evaluated and the concentration distribution of O_2 in the tube can be expressed as

$$\frac{C_A}{C_{A0}} = \frac{e^{v\,z/D_{AB}} - 1}{e^{v\,L/D_{AB}} - 1}$$

For very small values of $(v\ L)/D_{AB}$, the above equation becomes

$$\frac{C_A}{C_{AO}} = \frac{vz/D_{AB}}{vL/D_{AB}} = \frac{z}{L}$$

Problem

A model has to be developed for computing the distribution of NO_2 in the atmosphere. The molar flux of NO_2 at ground level, $N^{//}_{A,0}$ (as a result of automobile and other emissions) is known. The concentration of NO_2 at a distance well above ground level is zero and NO_2 reacts chemically in the atmosphere with unburned hydrocarbons to produce smog. The reaction is first order with rate expressed as $N_A = - k_1^{///} C_A$.

(i) Assume steady state and a diffusion only process to obtain an expression for the vertical distribution $C_A(x)$ of the molar concentration of NO_2 in the atmosphere.

(ii) If an NO_2 partial pressure of $p_A = 2\text{x}10^{-6}$ bar can cause pulmonary damage, what is the ground level molar flux for which a smog alert is to be issued? Data: $T = 300\text{K}$, $k_1 = 0.03\ s^{-1}$, $D_{AB} = 0.15\text{x}10^{-4}\ m^2/s$, $R = 8314\ \text{J/kmol. K}$

Species Balance Equation

$$\left(\cancel{\frac{\partial C_A}{\partial t}} + \cancel{v_x \frac{\partial C_A}{\partial x}} + \cancel{v_y \frac{\partial C_A}{\partial y}} + \cancel{v_z \frac{\partial C_A}{\partial z}}\right) = D_{AB}\left[\frac{\partial^2 C_A}{\partial x^2} + \cancel{\frac{\partial^2 C_A}{\partial y^2}} + \cancel{\frac{\partial^2 C_A}{\partial z^2}}\right] + R_A$$

Considerations

1. Steady state
2. Stagnant conditions, no convection
3. Homogeneous first order reaction
4. C_A is a function of x only
5. Molar flux at the ground level is known
6. Concentration of NO_2 is zero well above the ground level

Therefore, the governing equation is

$$\frac{d^2 C_A}{dx^2} - k'''_1 \, C_A = 0 \qquad (1)$$

$$C_A = C_1 \, e^{mx} + C_2 \, e^{-mx} \quad and \quad \frac{dC_A}{dx} = mC_1 \, e^{mx} - mC_2 \, e^{-mx} \qquad (2)$$

Where $m = \left(\frac{k'''_1}{D_{AB}}\right)^{1/2}$

Boundary Conditions

$$C_A\,(\infty) = 0 \qquad (3) \qquad \rightarrow \qquad C_1 = 0$$

$$\frac{dC_A}{dx}\Big|_{x=0} = -\frac{N''_{A0}}{D_{AB}} \qquad (4)$$

Known diffusive flux, N''_{A0} at x = 0

From Eq. (2) $\left.\frac{dC_A}{dx}\right|_{x=0} = -\,mC_2$ (5)

From Eq. (4) and (5) $-\,mC_2 = -\frac{N''_{A0}}{D_{AB}}$

Therefore From Eq. (2)

Vertical distribution of NO_2 in the atmosphere

$$\boxed{C_A = \frac{N''_{A0}}{mD_{AB}}\, e^{-mx}}$$

$$m = \left(\frac{k'''_1}{D_{AB}}\right)^{1/2}$$

$$m = \left(\frac{0.03}{0.15\ x\ 10^{-4}}\right)^{1/2} m^{-1} = 44.7\ m^{-1}$$

(ii) At the ground level, $C_A\,(0) = \frac{N''_{A0}}{(k'''_1\, D_{AB})^{1/2}}$

Partial pressure of NO_2 at the ground level, $p_A(0)$ is

$$p_A(0) = C_A(0)RT = \frac{RTN''_{A0}}{m\,D_{AB}} \quad and \quad C_{A0,Critical} = \frac{p_{A\,(0)Critical}}{RT}$$

$$C_{A0,Critical} = \frac{2\,x\,10^{-6} \times 10^5}{8314 \times 300} = 8 \times 10^{-8}\,\frac{k\,mol}{m^3}$$

$$R = 8314\,\frac{J}{K\,mol\,K} \qquad m = \left(\frac{0.03}{0.15\,x\,10^{-4}}\right)^{1/2} m^{-1} = 44.7\,m^{-1}$$

$$N''_{A0} = m\,D_{AB}\,C_A\,(0)$$

$$N''_{A0} = 44.7 \times 0.15 \times 10^{-4} \times 8 \times 10^{-8} = 5.37 \times 10^{-11}\,\frac{k\,mol}{m^2 s}$$

Problem

A spherical naphthalene ball of initial size of 0.03 m is hanged in a closet. One has to evaluate the time it takes for the naphthalene ball to sublimate completely. The following information are known. The concentration of naphthalene in the air is very small, and very low convective mass flux conditions exist inside the closet. Both the air and naphthalene vapor are ideal gases, the temperatures are same everywhere and there are no radiation effects. The molar mass of naphthalene is 128.2 kg/kmol with a density equal to 1100 kg/m^3 and

D_{AB} = 0.61x10^{-5} m^2/s. The dry air properties at 1 atm and 25^o C are ρ = 1.184 kg/m^3. c_p = 1007 J/Kg.K and α = 2.141 x 10^{-5} m^2/s, the mass fraction of naphthalene on the air side of the surface is 4.8 x 10^{-4}. You may use the appropriate value of Sherwood number for this case.

Evaluate the time required for the naphthalene ball to sublimate completely.

The mass fraction of N_2 in the air is very low and the ball is inside the closet, therefore there will hardly be any natural convection.

For such special cases, it has already been established (by equating the diffusive flus at the interface with the convective flux away from the interface) that the Sherwood number (as well as Nusselt number) has a constant value equal to 2

$$Sh = \frac{h_m D}{D_{AB}} = 2 \quad \rightarrow \quad h_m = \frac{2D_{AB}}{D}$$

The mass of the Naptahalene ball, $m = \rho_N \frac{1}{6} \pi D^3$

At the solid fluid surface of a sphere in stagnant systems

The diffusive heat (or mass) transfer at the interface denoted by $- k\ A\ \frac{dT}{dr}\Big|_{R\ =\ r}$ must be equal to the convective (hypothetical) heat (or mass) transfer away from the interface expressed as [h A (T_s - T_∞)] where h is the convective heat (or mass) transfer coefficient when the velocity approaches zero(stagnant condition).

From Eqn of energy, for a sphere in stagnant condition, at SS with no convection, T = f(r) only, T = T_s at r = R and T = T_∞ as r → ∞

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dT}{dr}\right) = 0$$

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dT}{dr}\right) = 0$$

Integrating and using the boundary conditions, the temperature distribution can be written as

$$\frac{T - T_\infty}{T_s - T_\infty} = \frac{R}{r}$$

$$\frac{dT}{dr} = -\frac{R}{r^2}(T_s - T_\infty) \qquad \frac{dT}{dr}\Big|_{r=R} = -\frac{1}{R}(T_s - T_\infty)$$

$$-k\frac{dT}{dr}\Big|_{r=R} = h(T_s - T_\infty) = -\frac{k}{R}(T_s - T_\infty) \rightarrow \frac{hR}{k} = 1$$

$$\frac{hD}{k} = 2 \rightarrow Nu = 2 \quad \text{and} \quad Sh = 2$$

The rate of change of the mass of N_2 is equal to the rate of mass transfer from N_2 to air. Therefore, (w being the mass fraction)

$$\frac{dm}{dt} = -\, h_m\, \rho_{Air}\, A\, (w_{NS} - w_{N\infty})$$

$$\frac{d}{dt}\left(\frac{1}{6}\, \rho_N \pi d^3\right) = -\, \frac{2\, D_{AB}}{D}\, \rho_{Air} \pi d^2 (w_{NS} - w_{N\infty})$$

$$\frac{\pi}{2}\, \rho_N\, D^2 \frac{dD}{dt} = -2\, D_{AB}\, \rho_{Air} \pi D (w_{NS} - w_{N\infty})$$

Integrating from D = D_i = 0.03 m at t = 0 to D = 0 at t = t (complete sublimation)

$$t = \frac{\rho_N {D_i}^2}{8\,\rho_{Air} D_{AB}(w_{NS} - w_{N\infty})} \qquad C_{N\infty} = 0$$

$$t = \frac{1100\ {}^{Kg}/_{m^3}\ (003)^2 m^2}{8\ \times 1.184\ {}^{Kg}/_{m^3} \times 0.61 \times 10^{-5}\ {}^{m^2}/_{s} \times 4.8 \times 10^{-4}}\ s$$

$$t = 3.57 \times 10^{7}\ s = 413\ days$$

A timed-release drug is dissolving in the intestine of a person. As a steady state approximation we may assume that the drug is a rod of overall radius r_o (m) and length L (m). The timed-release action is accomplished by putting an inert coating on the drug through which the drug diffuses with a diffusivity, D_{AB}. At the inner edge of the coating (r_i) the composition of the drug (mole fraction) is x_{ai}.

On the outer surface, the digestive juices provide for mass transfer with a mass transfer coefficient of k_{ac} (m/s) (equivalent to convective heat transfer coefficient). The amount of drug within the intestine can be approximated as $x_{a\infty} \approx 0$. Assume that the drug is not released by the two end-caps (circular) of the rod. You may also assume the total concentration of all species within the coating is c_t , a constant and it is a diffusion only process.

Calculate the rate of drug release (mg/hr) in the intestine. Given: r_i = 3mm, r_o = 5 mm, $D_{AB} = 10^{-10}\ m^2/s$, k_{ac} = 0.1 m/s, L = 5 mm, c_t = 0.4 kg/m^3, x_{ai} = 0.9.

Gov. Eqn. $$\frac{d}{dr}\left(r\,\frac{dx_A}{dr}\right) = 0$$

B.C.

i) Known concentration at r = r_i;
ii) Equality of diffusion and convection at r = r_o

The constants of integration can be evaluated

$$x_A = x_{A\,i} + \frac{k_{Ac}\,r_o}{D_{AB}}\;x_{Ao}\ln\frac{r_i}{r}$$

Putting the values $x_{Ao} = 3.527\ x\ 10^{-7}$

Rate of drug release into the intestine = Area x $k_{AC}\,[x_{Ao} - x_{A\infty}]\,C_t$

$$= 2\,\pi\,r_o\,L\,k_{AC}\,C_t\,x_{AO}$$

$= 8 \times 10^{-3}$ mg/hr

Heterogeneous Reaction

Nitric Oxide (NO) emissions from auto exhaust can be reduced by a catalytic Converter and the following reaction occurs at the catalyst surface, $NO + CO = 1/2N_2 + CO_2$.

The rate equation is given as. $r_A^{//} = -k_1^{//} C_A$ It may be assumed that NO reaches the catalyst surface by one-dimensional diffusion only process through a thin gas film of thickness L. The exhaust gases are at a temperature of 500 C and pressure of 1.2 bar. The thickness of the film is 1 mm and the reaction rate constant is, $k_1^{//} = 0.05\, m/s$

and the catalyst surface area is A = 200 cm^2. The diffusion coefficient D_{AB} = 10^{-4} m^2/s and the mole fraction of NO in the exhaust gases is $X_{AL} = 0.15$.

Evaluate the following

i) What is the mole fraction of NO at the catalyst surface?

ii) What is the NO removal rate for a surface area, A = 200 cm^2

$$\frac{d}{dz}\left(N_{Az}\right) = 0 \qquad \text{gov. eq}^{n}.$$

$$N_{Az} = -D_{AB}\frac{dc_A}{dz}$$

MOLE FRACTION

$$\frac{d^2 c_A}{dz^2} = 0 \qquad c\frac{d^2 x_A}{dz^2} = 0 \qquad \frac{d^2 x_A}{dz^2} = 0$$

$$x_A = c_1 z + c_2 \qquad c_1, c_2 \text{ CONST.}$$

I) B.C. $\quad z = L \quad x_A = x_{AL}$

II) B.C. $\quad -D_{AB}\cancel{c}\frac{dx_A}{dz} = -k_1''\cancel{c}\, x_{AS}$ EQUALITY OF DIFF^{N} RATE & REACT^{N} RATE

$$-D_{AB}c_1 = -k_1'' x_{AS}.$$

$$\boxed{c_1 = \frac{k_1''}{D_{AB}} x_{AS}} \qquad x_A = \frac{k_1'' x_{AS}}{D_{AB}} z + \underline{\underline{c_2}}.$$

$z = L \qquad x_A = x_{AL}$

$$x_{AL} = \frac{k_1'' x_{AS}}{D_{AB}} L + C_2$$

$$C_2 = x_{AL} - \frac{k_1'' x_{AS}}{D_{AB}} L.$$

$$x_A = \frac{k_1'' x_{AS}}{D_{AB}} z - \frac{k_1'' x_{AS}}{D_{AB}} L + x_{AL}.$$

$z = 0, \quad x_A = x_{AS}$ (SAY).

$$\therefore \boxed{x_{AS} = \frac{x_{AL}}{1 + \frac{k_1'' L}{D_{AB}}}} = \frac{0.15}{1 + \frac{0.001\,m \times 0.05\,\frac{m}{s}}{10^{-4}\,m^2/s}}$$

Part i) $\longrightarrow$ $\boxed{x_{AS} = 0.1}$

$$N''_{AS} = -k''_1 C_{AS} = -k''_1 C \cdot x_{AS}.$$

$$N''_{AS} = -\frac{k''_1 C \; x_{AL}}{1 + L k''_1 / D_{AB}}$$

$$C = \frac{p}{RT} = \frac{1.2 \text{ bar}}{8.314 \times 10^{-2} \frac{m^3 \cdot \text{bar}}{\text{kmol} \cdot K} \times 773\,K} = 0.0187 \frac{\text{kmol}}{m^3}$$

$$N''_{AS} = -9.35 \times 10^{-5} \frac{\text{kmol}}{m^2 \cdot s}.$$

MOLAR RATE OF NO REMOVAL FOR THE ENTIRE SURF.

$$N_{AS} = \underline{N''_{AS} \cdot A} = -9.35 \times 10^{-5} \frac{\text{kmol}}{m^2 \cdot s} \times 0.02\,m^2$$

Part ii)

$$\boxed{N_{AS} = -1.87 \times 10^{-6} \frac{\text{kmol}}{s}.}$$

Oxygen diffuses through the wall of drug containers and oxidizes many drugs rendering them inactive. It is the oxygen diffusion/reaction scenario that limits the shelf-life of many pharmaceutical products. To limit the oxidation of drugs, Oxygen scavengers, like sodium bisulfite ($NaHSO_3$) are often added to the drugs. The reaction to remove oxygen is:

$$2HSO_3^- + O_2 \rightarrow SO_4^{2-} + H_2O$$

Consider a liquid drug stored in a cylindrical polyethylene container. The container is 15 cm high and has an inner radius of 6 cm. The initial concentration of $NaHSO_3$ in the drug formulation is 1 g/lit. Neglect diffusion through the top and bottom of the container and assume that the $NaHSO_3$ reacts instantly with the O_2. The O_2 concentration in air is also always 21%; and this may be approximated as the concentration at the outer end of the container. The effective diffusivity of O2 through polyethylene is 9×10^{-13} m^2/s and you may assume P = 1 atm, T = 20 °C, and the process operates at steady-state. Evaluate the concentration profile of O_2 in the container material. How thick must the walls of the container be to ensure that 90% of the $NaHSO_3$ remains after 1 year?

Steady state diffusion, O2 conc. in drug = 0, O2 conc. in air = 21 %
Mol Wt. of SB = 104

1 D diffusion in cylindrical systems with no reaction

BC

i) known conc. At $r = r_0$

ii) Zero conc. At $r = r_i$ (instantaneous reacion)

Thickness 10 mm (approx)